User Manual

ST-MCA
Low-Cost STM32-Based Multi-Channel Analyzer

Saeed Boorboor

Project Manifesto

ST-MCA is a cost-effective Multi-Channel Analyzer (MCA) powered by the STM32x7 microcontroller series. Designed for high-performance signal processing, the system performs direct digitization of shaped signals and utilizes **Lagrange interpolation** to achieve high-precision peak detection.

The firmware supports versatile operational modes to accommodate various experimental setups, including:

- **Dual-Channel Mode:** Independent measurement of two separate detectors.
- **Coincidence Mode:** Real-time coincidence measurement of incoming pulses.
- **Correlation Analysis:** Recording and mapping channel data against one another for advanced statistical analysis.

Hardware Structure

The primary hardware stage consists of an amplifier circuit based on the AD8021 operational amplifier. The schematic for a single channel is illustrated in Figure 1. The input is designed to receive semi-Gaussian pulses from a 50-ohm standard shaping amplifier. Manual jumper switches are utilized to configure input signal polarity, and a selectable gain setting allows for amplification factors ranging from 1 to 20. To prevent low-voltage saturation at the ADC, an adjustable bias voltage of approximately 10 mV is applied to the OpAmp's positive terminal via resistor R34.

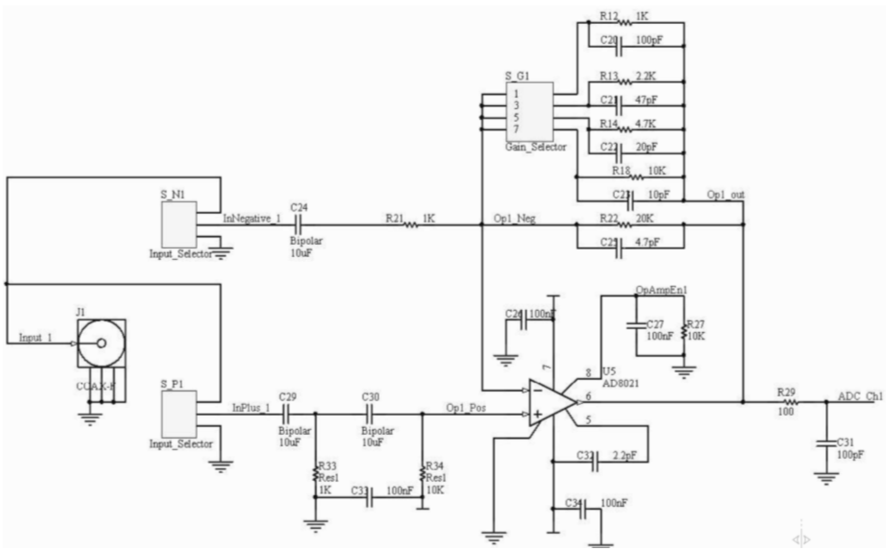

Figure 1: Schematic of the input amplifier.

The microcontroller connection schematic is illustrated in Figure 2. Three ADC channels are configured: two are interfaced with their respective input amplifiers, while the third is reserved for future high-sampling rate data acquisition requirements. Additionally, an LM75 temperature sensor (address: 0b000) is integrated into the system to monitor ambient conditions during spectral recording. Communication with the host computer is supported through both a virtual COM port (via USB) and a standard UART interface.

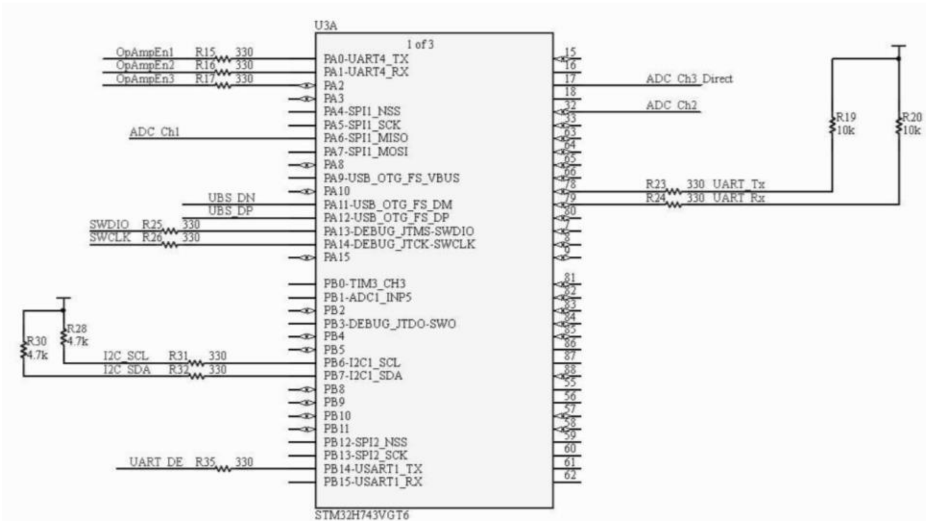

Figure 2: Hardware configuration of STM32H743 MCU.

The ST-MCA board layout is presented in Figure 3. The input stage on the left utilizes SMA connectors to receive analog signals from the shaping amplifier, with polarity selection jumpers positioned in close proximity to the inputs. Signal baseline adjustment is achieved using a multi-turn potentiometer to tap a portion of the reference voltage, which is subsequently buffered by an LM324 operational amplifier. To preserve signal integrity and minimize noise on the ADC inputs, digital signal lines are routed away from analog paths. The negative power supply for the operational amplifiers is derived from an isolated DC-DC converter located adjacent to the USB connector, which facilitates both board power and data communication.

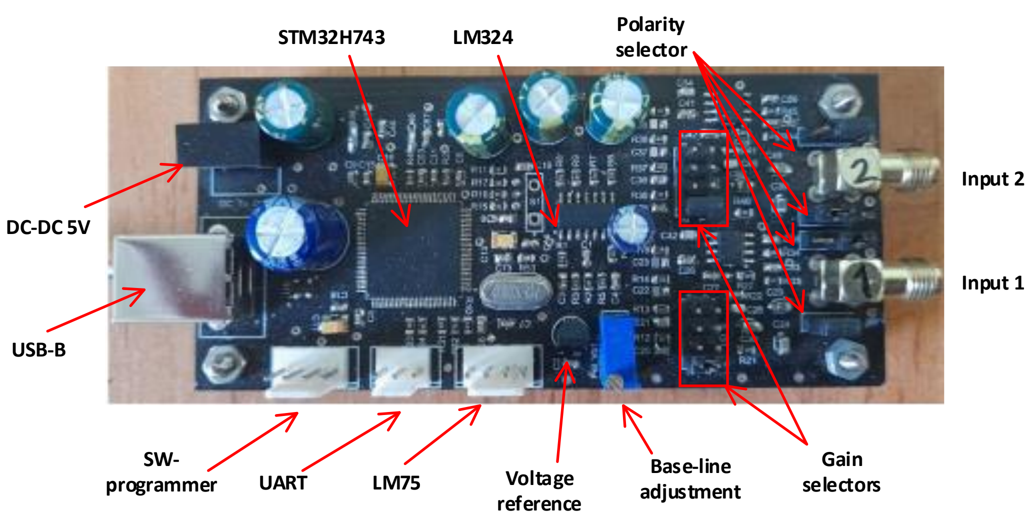

Figure 3: a photograph of the ST-MCA board.

Firmware structure

Figure 4 illustrates the firmware configuration of the MCA implemented in the STMH7. In this structure, a pseudo-Gaussian pulse is acquired as the input signal and sampled by the analog-to-digital converter (ADC) at a sampling rate of 3.6 MHz. Utilizing a Direct Memory Access (DMA) unit, these samples are automatically logged into a circular buffer, where writing continues from the beginning of the buffer once it becomes full. Concurrently, an analog watchdog unit continuously monitors the voltage level to detect the presence of incoming pulses within the sampled data. Upon pulse detection, the CPU executes mathematical algorithms for signal integration, pulse-height determination, baseline-shift correction, and related processing steps. The corrected pulse height is subsequently recorded in the pulse-height spectrum histogram. Upon receiving a spectrum read request from the host computer, the contents of the pulse-height spectrum buffer are transmitted to the computer and visualized within the device’s user-interface software.

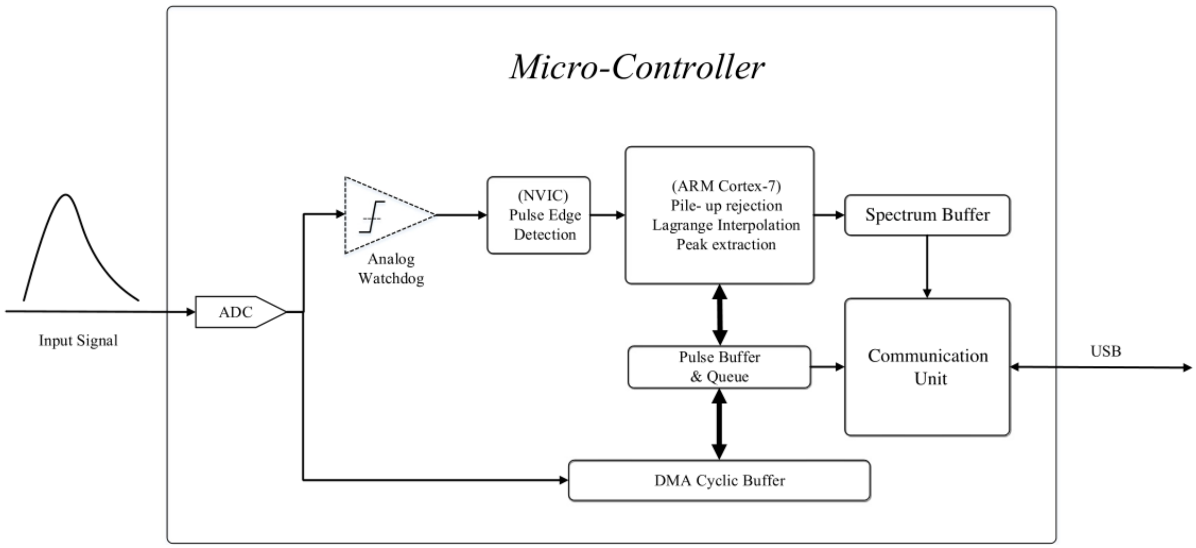

Figure 4: Firmware configuration of the MCA.

Spectrometer User Interface Section

To operate the instrument, a graphical user interface (GUI) was developed. The software was implemented in Python using the tkinter library, while the charting tools provided by matplotlib were utilized for visualizing the pulse-height spectrum.

Figure 5 depicts the main interface window of the device. This layout features a dedicated panel for displaying the pulse-height spectrum, equipped with built-in tools for plot zooming, panning, data export, and display configuration.

The left-hand panel provides control over the acquisition live/preset time. Additionally, a dedicated Start/Stop button facilitates the transmission of acquisition initiation and termination commands. The Report section displays the elapsed acquisition time alongside the real-time count rate of the spectrometer input channels.

The top menu bar contains two dropdown menus: File and Setting. As illustrated in Figure 6, the File menu enables operations such as initiating a new spectroscopy project and logging/saving spectrum data in CSV format. Meanwhile, the Setting menu allows the user to configure device communication parameters and verify hardware connectivity status.

Figure 7 shows a sample spectrum acquired from a Co-60 radioactive source over an acquisition time of 120 s using the developed system.

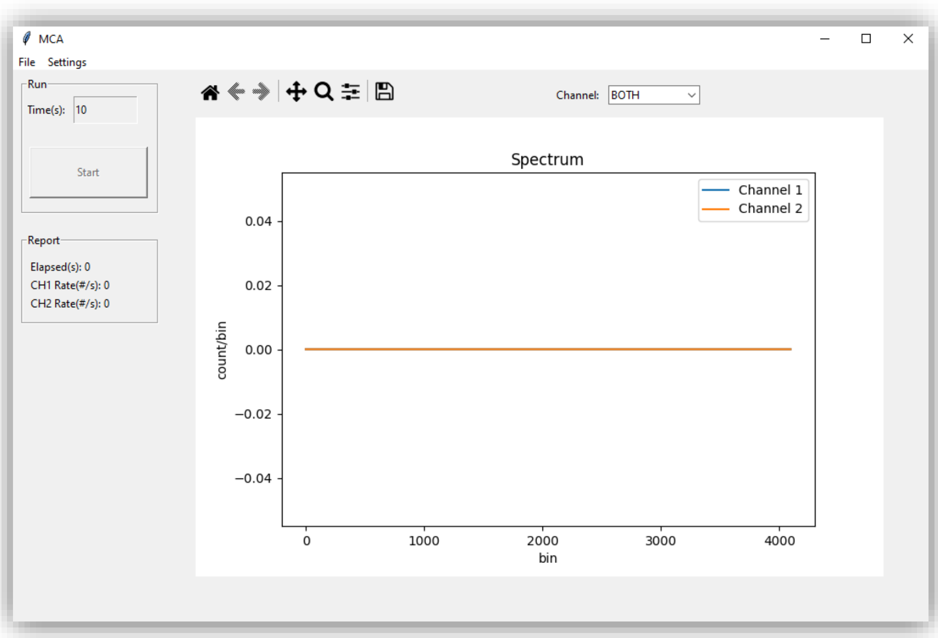

Figure 5: main interface window of the device.

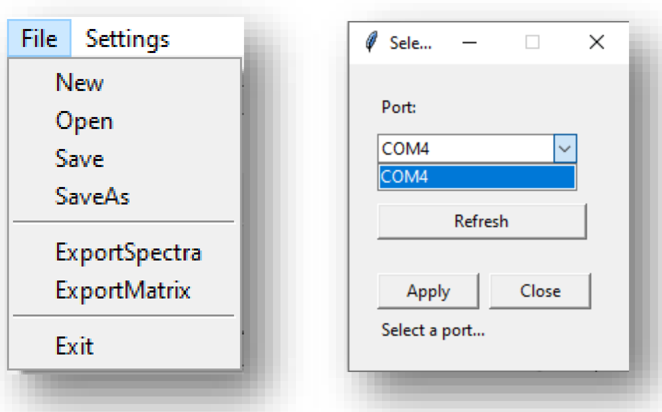

Figure 6: file and setting menus of the GUI.

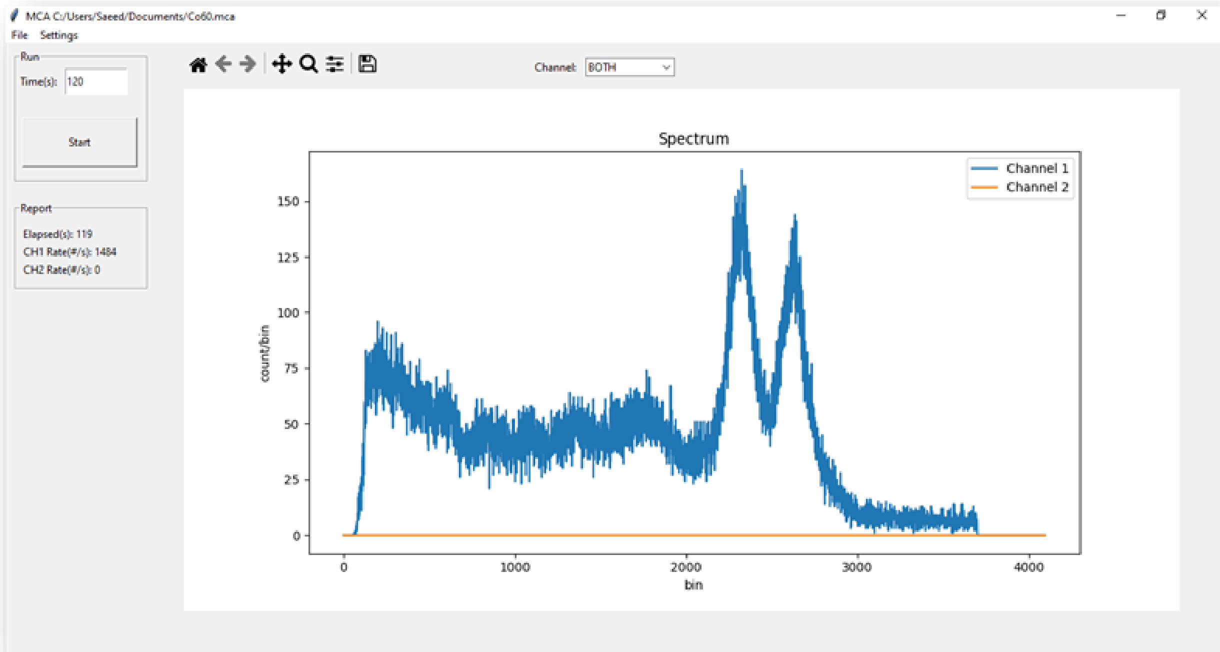

Figure 7: a representative Co-60 spectrum acquired by the device over a 120-second counting interval.